8. Reference fig. 7 and 8 - Installing switch.
Note: The HCP switch ships as a branch device with lugs located on the load end of the switch. To make switch a Main device, remove lugs from load side and install lugs on line side of switch and torque bolts to the values specified on the switch.
Position the switch so that the load side connectors are above and below the stab and the holes are aligned. Using the 5/16-18 hex head bolts (item 9), the 5/16" lock washers (item 16) and the 5/16" flat washers (item 10) attach the switch to the connectors using the inserts (item 24). Do not tighten. Repeat steps for all phases. Attach switch to support (item 12) using the two 1/4-20 X 1" machine screws (item 2) through the line side of the switch and into the support. Tighten screws. Tighten and torque the 5/16-18 bolts (item 9) to values shown on rear of deadfront.

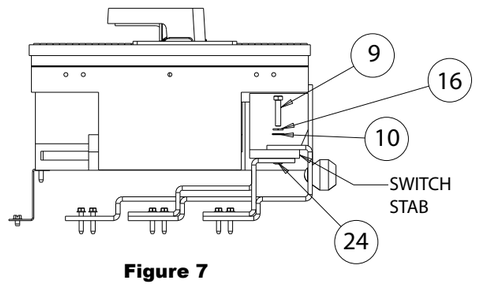


Figure 7

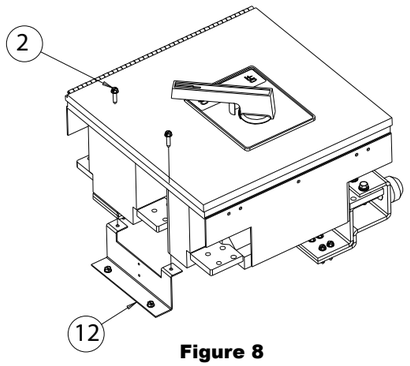


Figure 8

9. Reference fig. 9 - Installing ground assembly.
Orient the ring terminal on one end of the ground assembly (item 11) over the 0.171" hole in the switch main bracket (item 14). Orient the other ring terminal on the ground assembly over the 0.171" hole in the switch. Insert one 10-32 X 1/4" screw (item 4) through the ring terminal and into the 0.171" hole in the switch main bracket and tighten. Insert one 10-32 X 1/4" screw through the ring terminal and into the 0.171" hole in the switch and tighten.

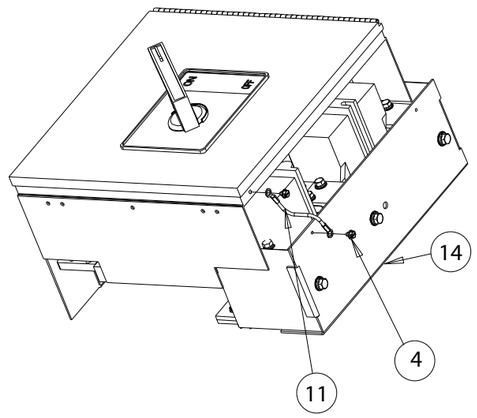


Figure 9

10. Re-install the deadfront and gutter covers.

SIEMENS

Power Panel Kit (P5) SMSKHCP for Main Mounting Type SENTRON HCP SWITCH

Installation Instructions

This device/system may only be set up and used in conjunction with these instructions. Installation, maintenance and operation of this device/system may only be performed by **qualified personnel.** Within the context of these instructions, qualified persons are defined as persons who have the skills and knowledge related to the construction and operation of the electrical equipment and installations and have received safety training to recognize and avoid the hazards involved.

11-D-1023-01 REV. 0

Installation Instructions

The following instructions are for the installation of Siemens Sentron HCP switches in a deep type panelboard (P5) as a Main. The parts provided in this kit are for connection to a 3 phase / 4 wire system.



1. Lock off power supplying this equipment before working on it.

2. Reference fig. 1 - Remove the (2) gutter covers, deadfront and all cover plates.

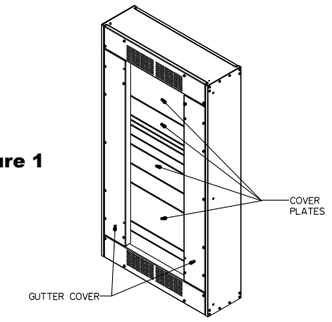


Figure 1

3. Reference fig. 2 - To locate mounting position of a HCP main switch, measure 15” from the top (or bottom for bottom feed applications) of Z-rail and mark position on Z-rail. The marked position acts as top (or bottom) of unit space for the strap assembly as shown in fig. 4.

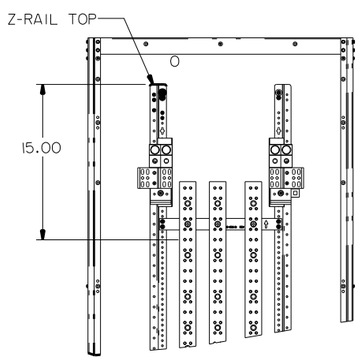


Figure 2

4. Open the shipping box and check the contents against the following list:

ITEM	DESCRIPTION	QTY.
1	1 3/8 STAND-OFF INSULATOR	3
2	1/4-20 X 1" SHWHSW	2
3	1/4-20 X 1-1/4" SHWHSW	12
4	10-32 X 1/4" SHWHSW	2
5	1/4-20 X 7/16" SHWHSW	4
6	3/8-16 X 3/4" CARRIAGE BOLT	3
7	3/8-16 X 3/4" HEX HEAD BOLT	3
8	3/8N WASHER	3
9	5/16-18 x 1-1/2" HEX HEAD BOLT	6
10	5/16N WASHER	6
11	GROUND STRAP ASSEMBLY	1
12	HCP BRACKET SUPPORT	1
13	HCP INSTALLATION INSTRUCTIONS	1
14	HCP SWITCH MAIN BRACKET	1
15	HELICAL SPRING LOCKWASHER 3/8"	3
16	HELICAL SPRING LOCKWASHER 5/16"	6
17	SUPPORT BARRIER	1
18	LONG PHASE CONNECTOR INNER	1
19	LONG PHASE CONNECTOR OUTER	1
20	MEDIUM PHASE CONNECTOR INNER	1
21	MEDIUM PHASE CONNECTOR OUTER	1
22	SHORT PHASE CONNECTOR INNER	1
23	SHORT PHASE CONNECTOR OUTER	1
24	THREADED INSERT	6

5. Reference fig. 3 - Installing inserts and stand-off insulators. Insert two threaded inserts (item 24) into the two 0.406” holes on the long phase connector inner (item 18) oriented as shown. (This is accomplished by tapping the inserts into the under-side of the connector with a hammer.) Do not strike hard enough to damage the threads. Repeat steps for medium (item 20) and short (item 22) phase connectors. Insert 3/8-16 X 3/4” carriage bolt (item 6) through the 0.437” square opening in the long phase connector outer (item 19) into the 1-3/8 stand-off insulator (item 1) and tighten. Repeat steps for medium (item 21) and short (item 23) phase connectors.

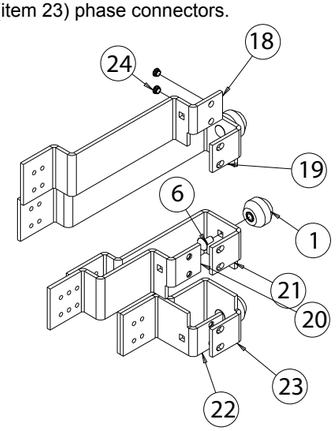


Figure 3

6. Reference fig. 4 - Attach phase straps and breaker support.
Locate the 0.228” diameter hole on the section bus 1.25” down from the top of the unit space established in step 3. Insert 1/4-20 X 1-1/4” screw (item 3) through the 0.312” diameter hole in the long phase connector inner (item 18), through the 0.312” diameter hole in the long phase connector outer (item 19) into the section bus. Hand tighten. Repeat for remaining three openings in long phase connectors. Locate the 0.228” diameter hole on the section bus 8.75” down from the top of the unit space. Insert 1/4-20 X 1-1/4” screw (item 3) through the 0.312” diameter hole in the medium phase connector inner (item 20), through the 0.312” diameter hole in the medium phase connector outer (item 21) into the section bus. Hand tighten. Repeat for remaining three openings in medium phase connectors. Locate the 0.228” diameter hole on the section bus 13.75” down from the top of the unit space. Insert 1/4-20 X 1-1/4” screw (item 3) through the 0.312” diameter hole in the short phase connector inner (item 22), through the 0.312” diameter hole in the short phase connector outer (item 23) into the section bus. Hand tighten. Repeat for remaining three openings in short phase connectors. Align the outside edge of the three phase connectors. Tighten and torque screws (item 3) to values shown on rear of dead front.
Locate the 0.228” diameter hole on the Z-rail 5.625” down from the top of the unit space. Insert 1/4-20 X 7/16” screw through the 0.312” hole in the HCP bracket support (item 12) into the 0.228” hole. Insert the second 1/4-20 X 7/16” (item 5) screw and tighten.

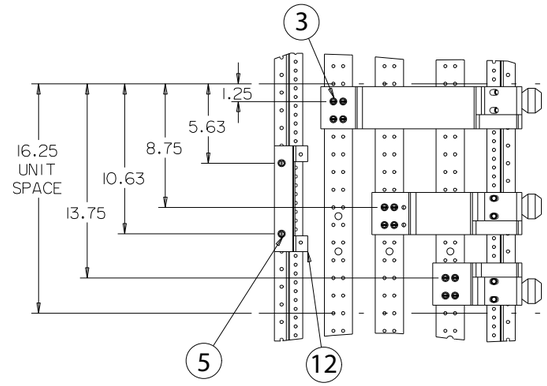


Figure 4
Note: orientation shown for left feed. For right feed, rotate 180 degrees.

7. Reference fig. 5, and fig. 6 - Installing support barrier assembly. Position the barrier (item 17) between the HCP bracket support (item 14) and the stand off insulators (item 1). Using the 3/8-16 hex head bolts (item 7), the 3/8” lock washers (item 15) and the 3/8” flat washers (item 8) attach insulation and support barrier to the stand off insulators.
Insert two 1/4-20 X 7/16” screws (item 5) through the HCP bracket (item 14) into the 0.228” diameter holes in the Z-rail. Tighten screws and bolts.

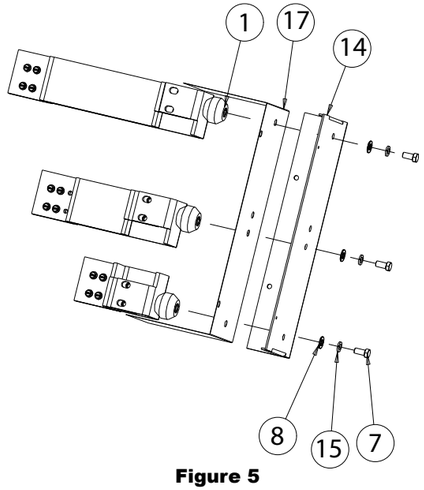


Figure 5

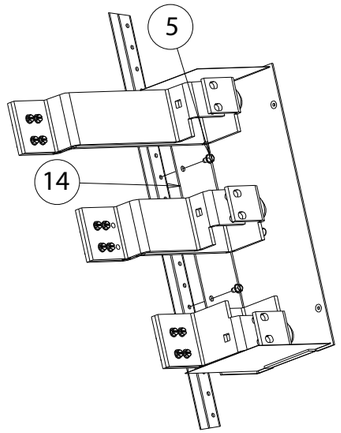


Figure 6